

Bash Linux Security Audit Automation

Repeatable host-check script for security posture review and evidence collection

Purpose

The included Bash script gathers commonly useful host-hardening evidence without changing the system. It checks system identity, listening ports, firewall status, SSH settings, UID 0 accounts, selected world-writable files, enabled services, disk usage, and recent account entries.

Check	Command family	Security value	Follow-up if abnormal
Listening ports	ss / netstat	Identifies exposed services and unexpected listeners.	Confirm owning process/business need; restrict or disable unnecessary service.
Firewall	ufw / firewall-cmd	Verifies host network filtering is active.	Apply default-deny and required service rules after change review.
SSH config	grep sshd_config	Reviews root login, password authentication, keys, retry limits.	Harden configuration and validate with sshd -T before restart.
UID 0 accounts	awk /etc/passwd	Finds accounts with full root UID.	Remove unauthorized UID 0 assignment; investigate account creation.
World-writable files	find -perm -0002	Finds writable files that may allow tampering.	Confirm business need; tighten owner/group/mode.
Enabled services	systemctl	Measures attack surface and startup persistence.	Disable unneeded service and validate application dependency.
Disk usage	df -h	Avoids failures that can disrupt logging/security tooling.	Alert before capacity affects logs or services.
Recent accounts	/etc/passwd review	Supports account-change review.	Correlate with authorized joiner/admin process.

Safety & Operational Design

- Read-only checks are used; the script does not automatically remediate findings.
- set -euo pipefail reduces silent script errors while individual optional commands are guarded.
- Tool availability is checked so the report remains readable across Linux distributions.
- Output is redirected to a timestampable report for review and evidence retention.
- Production use should add environment-specific configuration, permissions, logging, and change-control requirements.

Included Files

- bash/linux_security_audit.sh
- outputs/linux_security_audit_report.txt